MathVis Model

A decoder-only transformer trained FROM SCRATCH

★ MASTERS STANDOUT | Masters | Proud | Portfolio

Repo / Path	asadbek/AI ML/ai-edu/mathvis-model (frontend: mathvis-app)
Domain	Machine learning / NLP research
Stack	Python, PyTorch (from-scratch transformer), custom BPE tokenizer; Next.js frontend; FastAPI server
Status	Working research project - original ML work
Live / Deploy	Not publicly deployed

Overview

An original ~5M-parameter decoder-only transformer built and trained from scratch: 8 layers / 8 heads with SwiGLU activations and RoPE positional encoding, a custom 8192-vocab BPE tokenizer, a synthetic dataset generator, an AdamW + cosine training loop, an evaluation script and a FastAPI serving layer, with a Next.js frontend.

Key Features & Technical Highlights

- Modern transformer architecture implemented by hand: RoPE, SwiGLU, multi-head attention.
- Custom Byte-Pair-Encoding tokenizer (8192 vocab) written from scratch.
- Synthetic dataset generation + full training loop (AdamW, cosine schedule).
- Evaluation script and a FastAPI inference server; Next.js demo app.

Engineering Evidence & Metrics

- Genuine from-scratch model training - not fine-tuning or calling a pretrained API.
- End-to-end pipeline: tokenizer -> data -> train -> eval -> serve -> UI.

Why It Matters (application / interview talking points)

- The single strongest research signal in the portfolio for an ML/CS masters application.
- Shows you understand transformer internals deeply enough to reimplement them, not just use libraries.

Honest Caveats

- Lives inside the 'ai-edu' workspace alongside cloned reference repos - keep your original work clearly separated when presenting.